

Alyssa Kang

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EDUCATION

Harvard University

Candidate for Bachelor of Arts in Mathematics and Philosophy (GPA: 3.48)

Relevant Coursework: CS50, Math 22A, Math 55B, Math 117, Math 122, Math 124, Physics 15A, Stat 110

Stuyvesant High School

Advanced Regents Diploma, AP Scholar with Distinction (GPA: 96.3/100.0)

Cambridge, MA

Expected 2029

New York, NY

June 2025

RESEARCH & PROFESSIONAL EXPERIENCE

Langmuir Lab, Harvard Department of Earth and Planetary Sciences

Undergraduate Research Assistant

Cambridge, MA

January 2026 – Present

- Process deep-sea sediment cores for an NSF/ERC-funded study reconstructing up to 1.5M years of mid-ocean ridge volcanism: drying, decanting, acid-leaching carbonate, and isolating and mounting volcanic glass
- Measure major and trace element compositions of sediment-hosted basalt glasses via laser-ablation ICP-MS
- Build geochemical time series from glass data and apply statistical and spectral analysis to test whether glacial sea-level cycles modulate ocean-ridge magmatism

Measuring Mather

Research Assistant @ Mather House

Cambridge, MA

July 2026 – Present

- Design and run controlled experiments comparing bio-inspired (Nosehouse) and conventional (windcatcher/HVAC) ventilation, logging transient and steady-state temperature, humidity, and energy-consumption data
- Built a framework linking nasal morphology, entropy, and efficiency to identify exchanger designs for sustainable buildings

Cotera

Growth/GTM Intern

New York, NY

May 2026 – August 2026

- Restructured outbound outreach strategy, increasing response rates from 8% to 60% through refined targeting
- Streamlined website UI/UX and onboarding flow, simplifying navigation and reducing user time-on-task
- Engineered and shipped 5 application integrations, expanding product connectivity and functionality
- Produced multimedia storytelling and promotional materials, including product launch videos, across social media and digital channels in direct collaboration with the founder

Balyasny Asset Management

Macro Risk Management Intern

New York, NY

June 2025 – August 2025

- Managed a \$10M spot volatility risk portfolio including 21 macro assets, providing data-driven recommendations on optimal stock inclusion/allocation using linear algebra to maximize risk-adjusted returns
- Synthesized findings into a professional presentation and delivered results to external stakeholders

ENERGY & TECHNICAL PROJECTS

Harvard Undergraduate Clean Energy Group

Analyst, Energy Infrastructure Research for Last Energy

Cambridge, MA

January 2026 – May 2026

- Researched nuclear and traditional energy infrastructure to evaluate siting and market opportunities for Last Energy's modular nuclear power plants across candidate sites and customer segments
- Interviewed 20+ industry experts and stakeholders to inform deployment strategy and validate market assumptions
- Built techno-economic analyses of deployment costs; presented findings and recommendations to Last Energy's team

NucleAI | CS50 Final Project | github.com/ajkang/NucleAI

Fall 2026

- Built a full-stack AI web app (React, TypeScript, Express, OpenAI API) that helps nuclear startups draft NRC Safety Analysis Reports, structured around all 19 chapters of the NUREG-0800 Standard Review Plan
- Designed an end-to-end report workflow (AI-assisted drafting, section-level regeneration, progress tracking, formatted preview) on a type-safe schema layer shared across client and server; deployed live on Render

Blue Planet Alliance

Global Youth Ambassador

New York, NY

January 2022 – June 2025

- Implemented 100% renewable energy mandates in 20 US states and territories
- 1 of 100 youth delegates selected for the Local Conference of Youth; co-produced a 44-page National Youth Statement presented to Congress and COP29 and moderated UN events promoting 100% renewable energy

AWARDS, SKILLS, & INTERESTS

Awards: Prev. TeamUSA Table Tennis, Semifinalist @ NCTTA National Championships, NYS U19 Table Tennis Champion, United States Earth Science Olympiad, Terra Stem Fair Engineering Third Place Award, NYS GIS Applications Award

Skills and Certification: Mandarin (Fluent), Bloomberg Market Concepts (BMC), Python, TypeScript, React, ArcGIS, LA-ICP-MS

Interests: Smiling, pingin' & pongin', climbing rocks/minerals, and running through Central Park!